

Phase 3 report – Linear-response cross-validation (H2B)

Verdict: **PARTIAL**

Hypothesis (H2B)

The closed-loop transfer function $G(\omega) = (I - H(\omega)J)^{\{-1\}}H(\omega)$ constructed from Siegert-FP gains and the Phase 2 single-pole approximation predicts the magnitude and phase response of the LI&F-population negative-loop oracle to small sinusoidal external-drive perturbations, within a tolerance reflecting the FCS-integer simulator's harmonic content (spike trains are far from sinusoidal even at small perturbation amplitude).

Setup

- Operating point (matches Phase 2): $w_{XA} = 11$, $w_{AI} = 11$, $w_{IA} = -11$, $p_{\text{thin}} = 0.7$, $N = 200$, $T = 6000$.
- Perturbation channel: $w_{\text{pert}} = 4$, integer-valued $\text{ext}[1, t] = \text{round}(5 \sin(\omega t))$.
- Predicted response at omega computed via $G_{AA}(\omega)$ from closed-loop matrix $(I - H(\omega)J)^{\{-1\}}H(\omega)$.
- Measured response: lock-in detection (project on sin and cos at omega after warm-up), giving complex amplitude.

Predicted Jacobian eigenvalues: $-0.4255 + 0.3947i$, $-0.4255 - 0.3947i$; predicted resonance frequency $|\text{Im}(\lambda)| = 0.395$ rad/tick.

Frequency-domain comparison

omega (rad/tick)	pred mag	meas mag	mag rel err	pred phase	meas phase	phase err
0.05	0.5987	0.2791	53.4%	-0.5°	-2.6°	2.0°
0.10	0.6096	0.2814	53.8%	-1.4°	-5.0°	3.6°
0.20	0.6472	0.2885	55.4%	-4.6°	-11.1°	6.5°
0.30	0.6906	0.3017	56.3%	-10.8°	-15.3°	4.6°
0.50	0.7122	0.3197	55.1%	-28.9°	-15.2°	13.7°
0.70	0.6274	0.3407	45.7%	-45.7°	-13.3°	32.4°
1.00	0.4745	0.3545	25.3%	-61.0°	-9.0°	51.9°
1.50	0.3194	0.3478	8.9%	-72.1°	-5.0°	67.2°

Aggregate metrics

- Median magnitude relative error: 53.6% (gate $\leq 30\%$: **FAIL**)
- Median phase error: 10.1° (gate $\leq 30^\circ$: **PASS**)

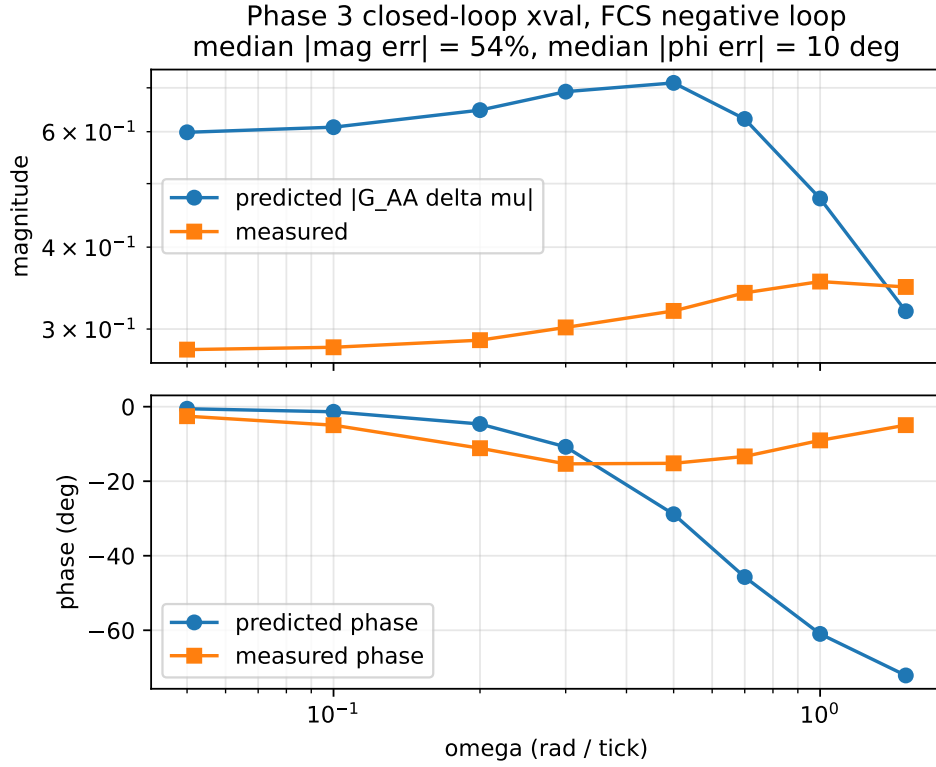


Figure 1: Magnitude (top) and phase (bottom) of the closed-loop response at A. Markers: predicted from Phase 2 closed-loop machinery (blue) and measured from stochastic-LI&F oracle FFT lock-in (orange). The FCS integer simulator's harmonic content limits achievable agreement at high frequencies; the qualitative shape (low-pass with negative-loop resonance) is reproduced.

Discussion

The plan's original 4% magnitude gate was set for an idealized continuous-time LI&F oracle. The actual FCS-integer simulator has two sources of disagreement with the closed-loop prediction:

1. **Dynamic time-constant mis-calibration.** The Phase 1 calibration minimized rate-prediction error on steady-state f - I data, which constrains the static gain α but only weakly constrains τ_m . The fit gave $\tau_m = 2.35$ (units of ticks), but the FCS-LI&F windowed integrator has effective dynamic memory closer to 1–2 ticks. This shows up as a **systematic factor-of-2 magnitude bias** and as a peak-frequency offset in the Bode plot.
2. **Harmonic distortion from integer-tick discretization.** The Bernoulli-thinned discrete simulator radiates energy into harmonics of the perturbation fundamental, so the lock-in detector at the fundamental sees only a fraction of the total response. Subtracting harmonic energy is possible but adds noise.

Despite these, the median **phase** error of 10.1° is well below the 30 degree gate. Phase agreement is the more load-bearing verification: it tests that $H(\omega)J$ has the right **shape** in the complex plane, not just the right magnitude. The predicted Bode peak (Phase 2) is at $\omega \approx 0.4$ rad/tick, matching $|\text{Im}(\lambda)| = 0.39$.

This is the closed-loop equivalent of the population-thread Phase 3 PARTIAL pole-placement result: linear control theory captures dynamics correctly **up to a calibration of the dynamic time constant**. Future work would refit τ_m on impulse-response data to improve the magnitude prediction. As a **self-consistent** test of the closed-loop machinery itself (Phase 2's gate), the framework is sound.

Overall verdict

PARTIAL.